

**Repeating Decimals to Fractions**Worksheet 7A.01.02 **ANSWER KEY****Use the repeating decimal 0.232323... for questions 1 through 4.**

1. Let
- $x = 0.232323\ldots$
- . Complete each equation.

$$10x = 2.\overline{32}$$

$$100x = 23.\overline{23}$$

Multiplying by 10 moves the decimal point one place:  $10 \times 0.232323\ldots = 2.323232\ldots$ . Multiplying by 100 moves it two places, which lines the repeating tails up with the original.

2. Which two values, when subtracted, leave a whole number? Fill in one bubble.

- ☐  $10x$  and  $x$
- ☒  **$100x$  and  $x$**
- ☐  $100x$  and  $10x$

Two digits repeat, so  $100x$  is the multiplier that lines the tails up.  $23.232323\ldots - 0.232323\ldots$  leaves a whole number.

3. Use two of the equations from question 1 to find the fraction equivalent of 0.232323... Show the subtraction step.

$$100x - x = 23.232323\ldots - 0.232323\ldots$$

$$99x = 23$$

$$x = \frac{23}{99}$$

$23/99$  will not simplify —  $99 = 9 \times 11$  and 23 is prime.

4. Write
- $8.\overline{23}$
- as a fraction and as a mixed number.

**Fraction:**  $\frac{815}{99}$

**Mixed number:**  $8\frac{23}{99}$

The decimal part is the  $23/99$  from question 3. Then  $8 \times 99 + 23 = 815$ .

**A trail mix recipe uses these amounts: raisins  $26.\overline{6}\%$ , peanuts  $3/8$ , chocolate chips 0.09, and the rest is chex mix.**

The bar means the 6 repeats forever.

5. What fraction of the recipe is chocolate chips?

$$\frac{9}{100}$$

0.09 has its last digit in the hundredths place.

6. What fraction of the recipe is raisins?

$$\frac{4}{15}$$
$$26.\overline{6}\% = 0.26666\dots \text{ Then } 100x - 10x \text{ gives } 90x = 24, \text{ so } x = 24/90 = 4/15.$$

7. What is the decimal equivalent of the peanuts?

$$0.375$$
$$3 \div 8 = 0.375 \text{ exactly.}$$

8. About what percent of the recipe is chex mix? Round to the nearest whole percent.

$$27\%$$

Over 600ths:  $4/15 = 160/600$ ,  $3/8 = 225/600$  and  $9/100 = 54/600$ , which total  $439/600$ . Then  $600 - 439 = 161$ , and  $161 \div 600 = 0.26833\dots$ , which rounds to 27%.

For questions 9 and 10, complete the table.

|     | Decimal             | Percent               | Fraction or Mixed Number |
|-----|---------------------|-----------------------|--------------------------|
| 9.  | $4.\overline{65}$   | $465.\overline{65}\%$ | $4\frac{65}{99}$         |
| 10. | $0.06\overline{24}$ | $6.\overline{24}\%$   | $\frac{103}{1650}$       |

9) Move the decimal two places right for the percent. For the fraction,  $100x - x$  gives  $99x = 461$ , so  $x = 461/99 = 4\frac{65}{99}$ . 10) Clear the repeat first:  $100x - x$  on  $6.242424\dots$  gives  $99x = 618$ , so  $6.242424\dots = 618/99 = 206/33$ . A percent is out of 100, so divide by 100 at the end:  $206/3300 = 103/1650$ .